Computer Vision:

Week-6 (23-27 Feb) Theory Lectures

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Review

- Segmentation (Isolation part)
- Shape Analysis (Description part)
- Advanced structural descriptors
- Circularity
- Aspect ratio

Segmentation

Dividing an image into meaningful parts (objects/regions).

- Original Image → Contains cat + background
- Segmented Image → Only cat is separated

Segmentation = Find object boundaries and regions of interest (ROI)

Edge-Based Segmentation

- *Instead of separating by color or region, we separate using edges (boundaries).*

Edge = Sudden change in intensity (brightness)

Black background + White object
Boundary between them = Edge

Contour Processing (Very Important in Shape Analysis)

Contour = Boundary outline of an object.

Contour = Shape border of an object

$$Circle \rightarrow Circular\ contour$$

$$Rectangle \rightarrow Rectangular\ contour$$

Connected Components

Connected Components = Group of connected pixels that belong to the same object.

White pixels = Objects

Black pixels = Background

All connected white pixels = One component (one object)

Why Connected Components Are Important

Counting objects in image

License plate detection

Medical cell counting

ROI extraction

Shape Descriptors (Core of Shape Analysis)

Shape Descriptors = Numerical features that describe the shape of an object.

Area
Number of pixels inside object

Perimeter
Length of object boundary

Bounding Box
Rectangle covering object

Method	*Purpose*	*Best Use*
Edge-Based Segmentation	Detect boundaries	Object borders
Contours	Extract shape outline	Shape analysis
Connected Components	Detect separate objects	Object counting
Shape Descriptors	Describe shape numerically	ML & classification

Machine Learning for Vision

Machine Learning for Vision **=** *Using algorithms that learn from images/videos to automatically understand visual data.*

- Face recognition
- Tumor detection
- Object detection
- Self-driving cars vision

Computer Vision + Machine Learning

- **Computer Vision**

Gives the computer the ability to **process images**

- **Machine Learning**

Gives the computer the ability to **learn patterns from images**

- **Combined:**

Image → Features → ML Model → Prediction

Types of Machine Learning Used in Vision

Supervised Learning (Most Common)

Model learns from **labeled images**

- *Input: Cat image (Label = Cat)*
- *Output: Model learns cat features*

Types of Machine Learning Used in Vision

Supervised learning

Common Algorithms

- *Support Vector Machine (SVM)*
- *K-Nearest Neighbors (KNN)*
- *Random Forest*
- *Neural Networks*

Types of Machine Learning Used in Vision

Unsupervised Learning

Idea:

Model learns patterns **without labels**

Example:

- Image clustering
- Image segmentation

Algorithms:

- K-Means Clustering
- PCA (Dimensionality reduction)

Deep Learning (Advanced Vision ML)

Most powerful method today.

Uses:

- CNN (Convolutional Neural Networks)
- R-CNN, YOLO, Vision Transformers

Image Feature Vector Size

*-An RGB image of size **64 × 64** is used for machine learning classification.*

If the image is flattened into a feature vector, What should be :

- *Total number of features*
- *Feature vector dimension*

Accuracy Calculation

-A vision model classifies 200 images:

- Correctly classified = 170
- Incorrectly classified = 30

Accuracy Calculation

$$\text{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}} \times 100$$

Confusion Matrix

$$A\ binary\ image\ classifier\ \left(Cat\ vs\ Dog\right)\ gives$$

$$:True\ Positive\ \left(TP\right)\ =\ 50$$

$$True\ Negative\ \left(TN\right)\ =\ 40$$

$$False\ Positive\ \left(FP\right)\ =\ 5$$

$$False\ Negative\ \left(FN\right)\ =\ 5$$

Confusion Matrix

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Machine Learning for Vision

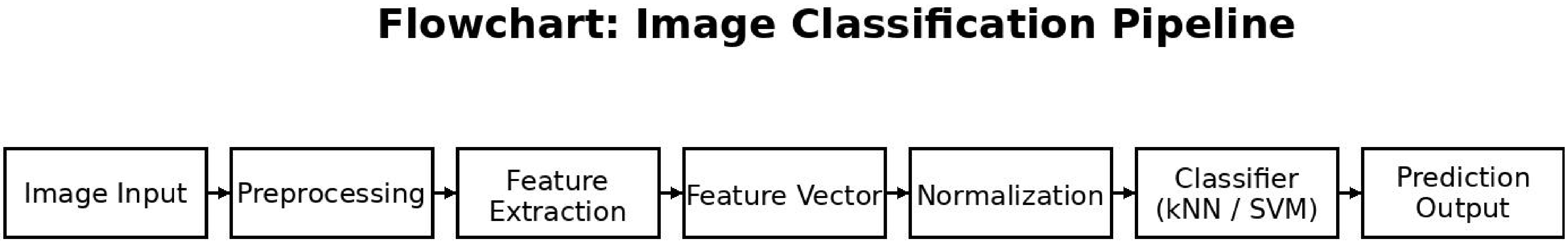
What is Machine Learning in Computer Vision?

Machine Learning for Vision

Feature Vectors

- **Normalization**
- **Traditional Classifiers**
    - k-Nearest Neighbors (kNN)
    - Support Vector Machine (SVM)

Continue


Flowchart: Image Classification Pipeline
Image Input
Preprocessing
Feature Extraction
Feature Vector
Normalization
Classifier (kNN / SVM)
Prediction Output

Continue

Feature Vector in Computer Vision

- A feature vector is a numerical representation of an image.

Instead of using full images, we convert them into:

- Color features (RGB values)
- Texture features (LBP, Gabor)
- Shape features (edges, contours)
- Deep features (CNN embeddings)

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Types of Visual Features

Common Feature Types in Vision

1. Color Features
   - Histogram, RGB values
2. Texture Features
   - LBP, GLCM
3. Shape Features
   - Contours, edges, blobs
4. Deep Features
   - Extracted from CNN models

Continue

Feature Normalization

Why Normalization is Important?

Feature normalization scales data into a common range.

Without normalization:

- Some features dominate others
- Poor classifier performance
- Slow convergence

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Common Methods:

- Min-Max Normalization
- Z-score Standardization
- L2 Normalization

Continue

$$The\ Min-Max\ Normalization\ Formula$$

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Continue

Status	*Vector Values*
Raw (Before)	[100, 5, 200]
Scaled (After)	[0.5, 0.02, 1.0]

Continue

k-Nearest Neighbors (kNN) in Vision

kNN Classifier for Image Recognition

kNN is a distance-based classifier.

How it Works:

1. Store training feature vectors
2. Calculate distance (Euclidean)
3. Find k nearest samples
4. Assign majority class

Continue

Distance Formula:

$$d = \sqrt{\sum_{i=1}^{n} (x_i - y_i)^2}$$

Continue

Support Vector Machine (SVM) in Vision

SVM for Image Classification

SVM finds the optimal hyperplane that separates classes.

Key Advantages:

- Works well with high-dimensional image features
- Robust to noise
- Effective for small datasets

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Kernel Types:

- Linear Kernel
- RBF Kernel
- Polynomial Kernel

<u>*Example Applications:*</u>

- Tumor classification (MRI)
- Object recognition

Continue

Kernel	*Shape of Boundary*	*Best for*
Linear	Straight line	Simple, linearly separable
RBF	Curved / circular	Complex, non-linear
Polynomial	Polynomial curves	Moderate complexity, patterns

Continue

Feature	kNN	SVM
Type	Lazy Learning	Supervised Learning
Speed	Slow (testing)	Fast (after training)
Accuracy	Moderate	High
Best For	Small datasets	Complex vision tasks

Continue

- Feature vectors convert images into numerical form
- Normalization improves model performance
- kNN is simple and intuitive for vision tasks
- SVM is powerful for high -dimensional image data

Simple Feature Vector Creation

$$I = \begin{bmatrix} 10 & 20 \\ 30 & 40 \end{bmatrix}$$

Min-Max Normalization

$By\ applying\ the\ Min - Max\ Normalization\ formula\ to\ the\ feature\ vector$

$[10,\ 20,\ 30,\ 40],$

how to $scale\ the\ data\ to\ the\ range [0,\ 1].$

$$X_{norm} = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Min-Max Normalization

$$The\ Extracted\ Vector$$

$$\mathbf{V\_\{norm\}\ =\ [0,\ 0.333,\ 0.667,\ 1]}$$

Z-Score Normalization

Feature vector= $[2,4,6,8]$

$$Z = \frac{x_i - \mu}{\sigma}$$

Mean= $$\mu = \frac{\sum_{i=1}^{N} x_i}{N}$$

Std= $$\sigma = \sqrt{\frac{\sum_{i=1}^{N} (x_i - \mu)^2}{N}}$$

Z-Score Normalization

$$Z = [-1.34, -0.45, 0.45, 1.34]$$